

SALES DATA ANALYSIS AND REPORTING FOR A RETAIL CHAIN

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
print("Libraries imported Successfully")
```

Libraries imported Successfully

1. LOAD DATASET

```
df = pd.read_csv('/content/OnlineRetail.csv',encoding='latin1')

print("Dataset Loaded Successfully")
```

Dataset Loaded Successfully

```
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom	
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
25010	538351	22149	FELTCRAFT 6 FLOWER FRIENDS	23	12/10/2010 15:17	5.06	NaN	United Kingdom	
25011	538351	22150	3 STRIPEY MICE FELTCRAFT	1	12/10/2010 15:17	5.06	NaN	United Kingdom	
25012	538351	22174	PHOTO CUBE	4	12/10/2010 15:17	4.21	NaN	United Kingdom	

```
df.sample(10)
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
188597	553048	23080	RED METAL BOX TOP SECRET	2	5/13/2011 10:02	8.25	14291.0	United Kingdom	
175432	551958	20675	BLUE POLKADOT BOWL	8	5/5/2011 12:37	1.25	13883.0	United Kingdom	
431748	573748	84879	ASSORTED COLOUR BIRD ORNAMENT	24	11/1/2011 9:24	1.69	14911.0	EIRE	
73324	542344	22678	FRENCH BLUE METAL DOOR SIGN 3	1	1/27/2011 11:30	2.46	NaN	United Kingdom	
376371	569523	23321	SMALL WHITE HEART OF WICKER	2	10/4/2011 14:41	1.65	16033.0	United Kingdom	
233896	557494	22488	NATURAL SLATE RECTANGLE CHALKBOARD	3	6/20/2011 15:19	3.29	NaN	United Kingdom	
.....	-----	-----	-----	--	11/22/2011	- - -	-----	United	

2. DATA UNDERSTANDING

df.shape

(541909, 8)

df.columns

Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate', 'UnitPrice', 'CustomerID', 'Country'], dtype='object')

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
Column Non-Null Count Dtype
--- ---
0 InvoiceNo 541909 non-null object
1 StockCode 541909 non-null object
2 Description 540455 non-null object
3 Quantity 541909 non-null int64
4 InvoiceDate 541909 non-null object
5 UnitPrice 541909 non-null float64
6 CustomerID 406829 non-null float64
7 Country 541909 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB

df.dtypes

0
InvoiceNo object
StockCode object
Description object
Quantity int64
InvoiceDate object
UnitPrice float64
CustomerID float64
Country object

dtype: object

df.describe()

	Quantity	UnitPrice	CustomerID	
count	541909.000000	541909.000000	406829.000000	
mean	9.552250	4.611114	15287.690570	
std	218.081158	96.759853	1713.600303	
min	-80995.000000	-11062.060000	12346.000000	
25%	1.000000	1.250000	13953.000000	
50%	3.000000	2.080000	15152.000000	
75%	10.000000	4.130000	16791.000000	
max	80995.000000	38970.000000	18287.000000	

df.describe(include='object')

	InvoiceNo	StockCode	Description	InvoiceDate	Country	
count	541909	541909	540455	541909	541909	
unique	25900	4070	4223	23260	38	
top	573585	85123A	WHITE HANGING HEART T-LIGHT HOLDER	10/31/2011 14:41	United Kingdom	
freq	1114	2313	2369	1114	495478	

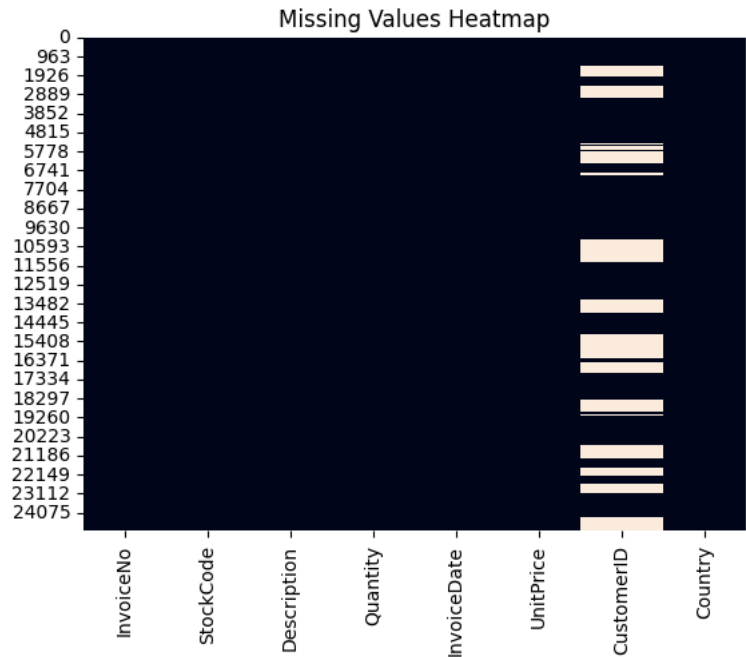
3. MISSING VALUE ANALYSIS

df.isnull().sum()	
	0
InvoiceNo	0
StockCode	0
Description	111
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	8959
Country	0
dtype: int64	

(df.isnull().sum()/len(df))*100	
	0
InvoiceNo	0.000000
StockCode	0.000000
Description	0.443734
Quantity	0.000000
InvoiceDate	0.000000
UnitPrice	0.000000
CustomerID	35.814511
Country	0.000000
dtype: float64	

(df.isnull().sum()/len(df))*100	
	0
InvoiceNo	0.000000
StockCode	0.000000
Description	0.443734
Quantity	0.000000
InvoiceDate	0.000000
UnitPrice	0.000000
CustomerID	35.814511
Country	0.000000
dtype: float64	

```
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values Heatmap")
plt.show()
```



```
df[df["CustomerID"].isnull()].head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
	622	536414	22139	NaN	56	12/1/2010 11:52	0.00	NaN	United Kingdom
	1443	536544	21773	DECORATIVE ROSE BATHROOM BOTTLE	1	12/1/2010 14:32	2.51	NaN	United Kingdom
	1444	536544	21774	DECORATIVE CATS BATHROOM BOTTLE	2	12/1/2010 14:32	2.51	NaN	United Kingdom

4. DUPLICATE ANALYSIS

```
df.duplicated().sum()
```

```
np.int64(352)
```

```
df[df.duplicated()].head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
	517	536409	21866	UNION JACK FLAG LUGGAGE TAG	1	12/1/2010 11:45	1.25	17908.0	United Kingdom
	527	536409	22866	HAND WARMER SCOTTY DOG DESIGN	1	12/1/2010 11:45	2.10	17908.0	United Kingdom
	537	536409	22900	SET 2 TEA TOWELS I LOVE LONDON	1	12/1/2010 11:45	2.95	17908.0	United Kingdom

5. DATA CLEANSING

```
df = df.dropna(subset=["Description"])
```

```
df = df.drop_duplicates()
```

```
df.shape
```

(536641, 8)

6. FEATURE ENGINEERING

```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]  
  
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom	15.30	
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom	20.34	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom	22.00	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
```

```
df["Year"] = df["InvoiceDate"].dt.year  
df["Year"].head()
```

	Year
0	2010
1	2010
2	2010
3	2010
4	2010

dtype: int32

```
df["Day"] = df["InvoiceDate"].dt.day_name()  
df["Day"].head()
```

	Day
0	Wednesday
1	Wednesday
2	Wednesday
3	Wednesday
4	Wednesday

dtype: object

7. BASIC KPI ANALYSIS

```
df["Revenue"].sum()  
  
np.float64(419656.89999999997)
```

```
df['Revenue'].mean()  
  
np.float64(17.092574942978167)
```

```
df['Revenue'].median()  
  
8.299999999999999
```

```
df['Revenue'].min()
```

-13541.33

df['Revenue'].max()

13541.33

8. PRODUCT ANALYSIS

df["Description"].nunique()

2478

df.groupby("Description")["Revenue"].sum().sort_values(ascending=False).head(10)

	Revenue
Description	
REGENCY CAKESTAND 3 TIER	16883.77
DOTCOM POSTAGE	14558.26
VINTAGE UNION JACK MEMOBOARD	6701.13
PAPER CHAIN KIT 50'S CHRISTMAS	6237.48
WHITE HANGING HEART T-LIGHT HOLDER	6073.86
WOOD BLACK BOARD ANT WHITE FINISH	6032.99
CREAM HEART CARD HOLDER	5280.32
BLACK HEART CARD HOLDER	5085.02
HOT WATER BOTTLE TEA AND SYMPATHY	4932.88
CHILLI LIGHTS	4706.32

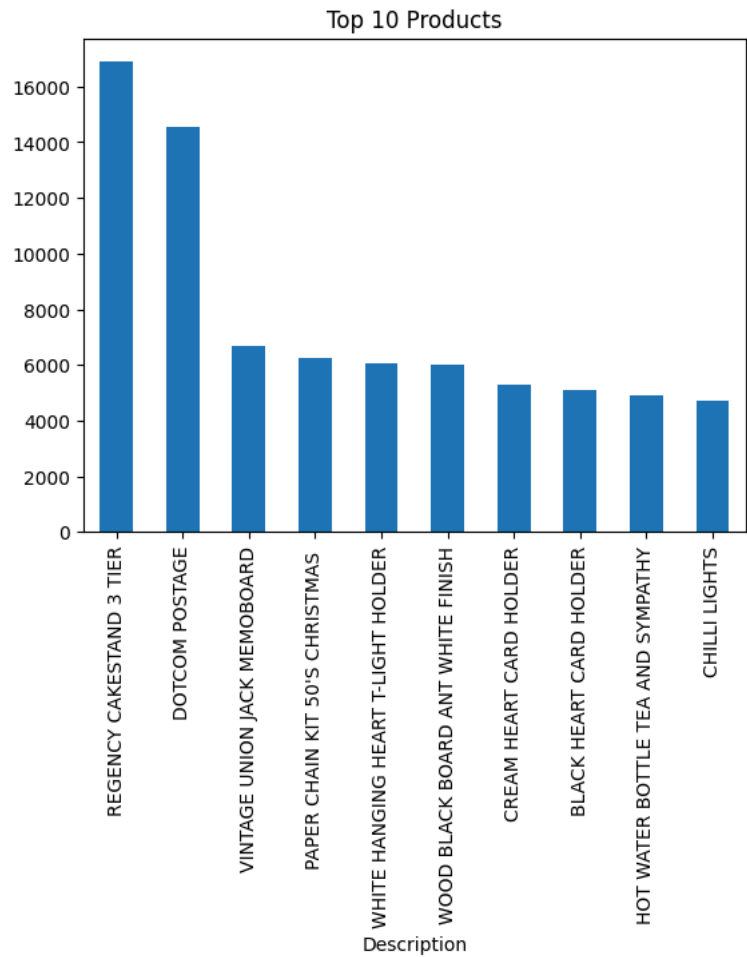
dtype: float64

df.groupby("Description")["Quantity"].sum().sort_values(ascending=False).head(10)

	Quantity
Description	
WORLD WAR 2 GLIDERS ASSTD DESIGNS	3826
CREAM HEART CARD HOLDER	2168
BLACK HEART CARD HOLDER	2130
MINI PAINT SET VINTAGE	2122
WHITE HANGING HEART T-LIGHT HOLDER	2068
PACK OF 72 RETROSPOT CAKE CASES	1930
ASSORTED COLOUR T-LIGHT HOLDER	1884
RED HARMONICA IN BOX	1821
ASSORTED COLOUR BIRD ORNAMENT	1660
PAPER CHAIN KIT 50'S CHRISTMAS	1658

dtype: int64

df.groupby("Description")["Revenue"].sum().sort_values(ascending=False).head(10).plot(kind="bar")
plt.title("Top 10 Products")
plt.show()



9. COUNTRY ANALYSIS

```
df["Country"].nunique()
```

19

```
df["Country"].value_counts().head(10)
```

count	
Country	
United Kingdom	23247
Germany	308
France	287
EIRE	228
Norway	147
Portugal	94
Spain	45
Lithuania	35
Iceland	31
Italy	25

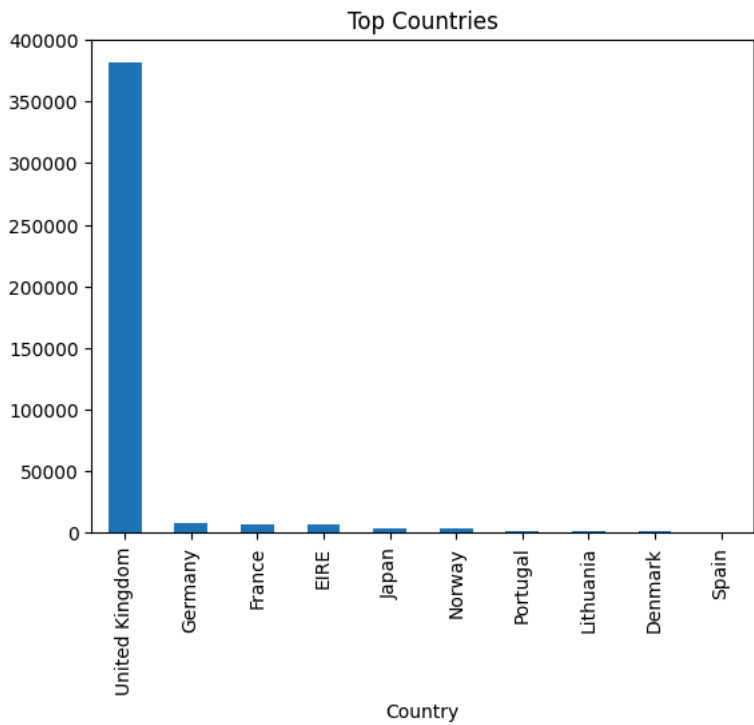
dtype: int64

```
df.groupby("Country")["Revenue"].sum().sort_values(ascending=False).head(10)
```

Revenue	
Country	
United Kingdom	381272.24
Germany	7795.74
France	7335.17
EIRE	6431.33
Japan	4114.48
Norway	3787.12
Portugal	1976.47
Lithuania	1661.06
Denmark	1281.50
Spain	794.72

dtype: float64

```
df.groupby("Country")["Revenue"].sum().sort_values(ascending=False).head(10).plot(kind="bar")
plt.title("Top Countries")
plt.show()
```



10. CUSTOMER ANALYSIS

```
df["CustomerID"].nunique()
```

654

```
df.groupby("CustomerID")["Revenue"].sum().sort_values(ascending=False).head(10)
```


Revenue	
CustomerID	
18102.0	27834.61
15061.0	17344.78
13777.0	6798.16
14911.0	5567.63
17850.0	5391.21
16210.0	4738.54
16029.0	4271.52
12755.0	3794.40
12433.0	3787.12
17381.0	3603.72

dtype: float64

11. TIME SERIES ANALYSIS

```
df.groupby("Year")["Revenue"].sum()
```

Revenue	
Year	
2010	419656.9

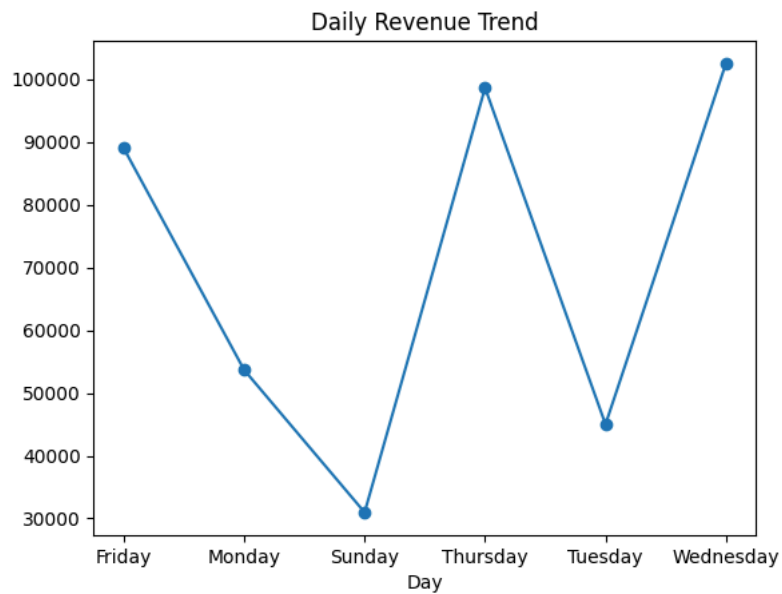
dtype: float64

```
df.groupby("Day")["Revenue"].sum()
```

Revenue	
Day	
Friday	88965.46
Monday	53653.87
Sunday	30973.63
Thursday	98582.46
Tuesday	44994.70
Wednesday	102486.78

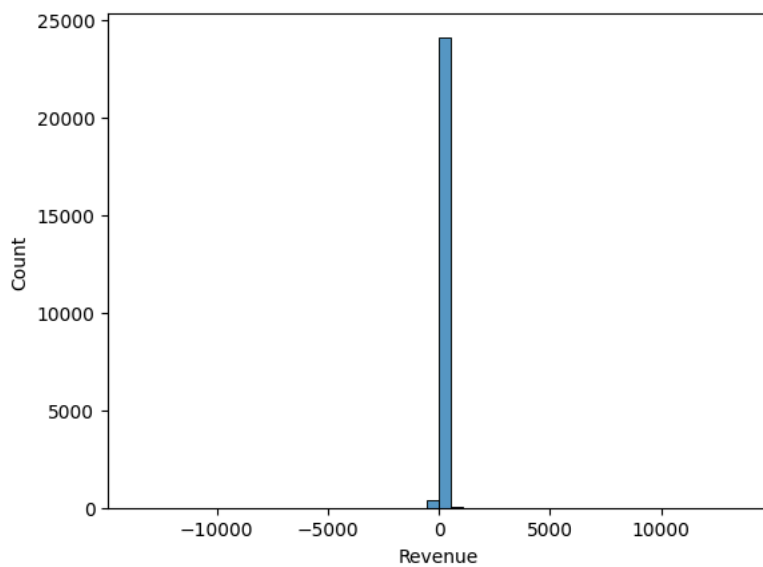
dtype: float64

```
df.groupby("Day")["Revenue"].sum().plot(marker="o")
plt.title("Daily Revenue Trend")
plt.show()
```

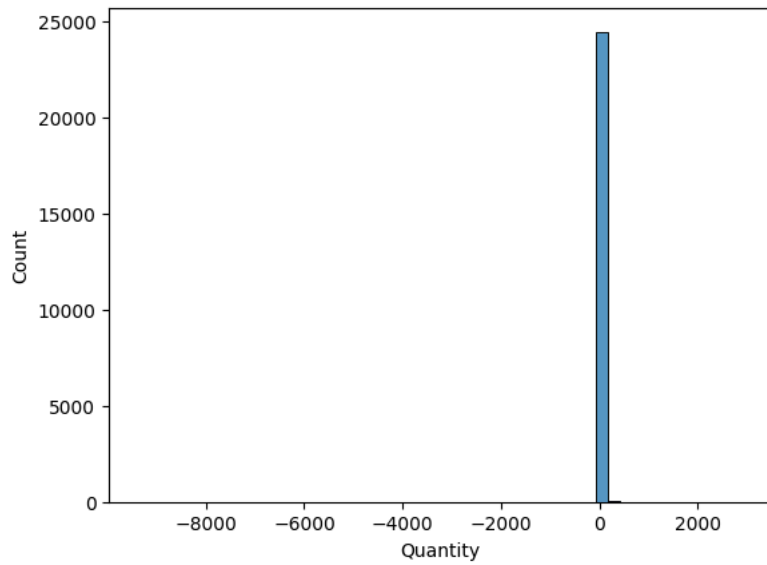


12. DISTRIBUTION ANALYSIS

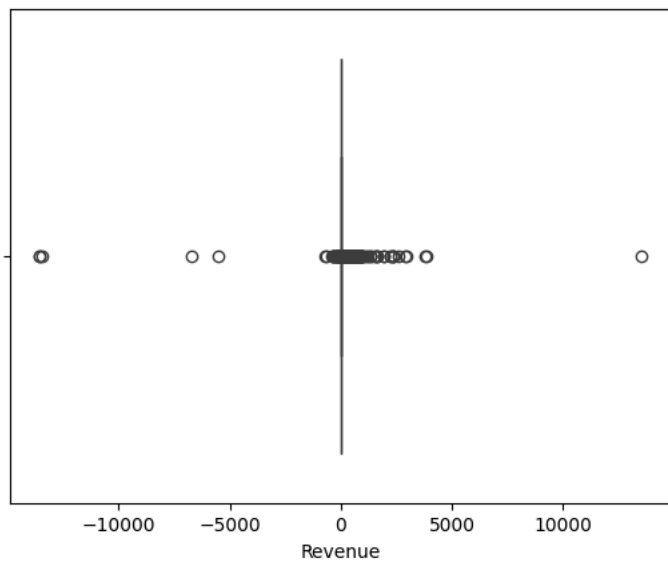
```
sns.histplot(df["Revenue"], bins=50)  
plt.show()
```



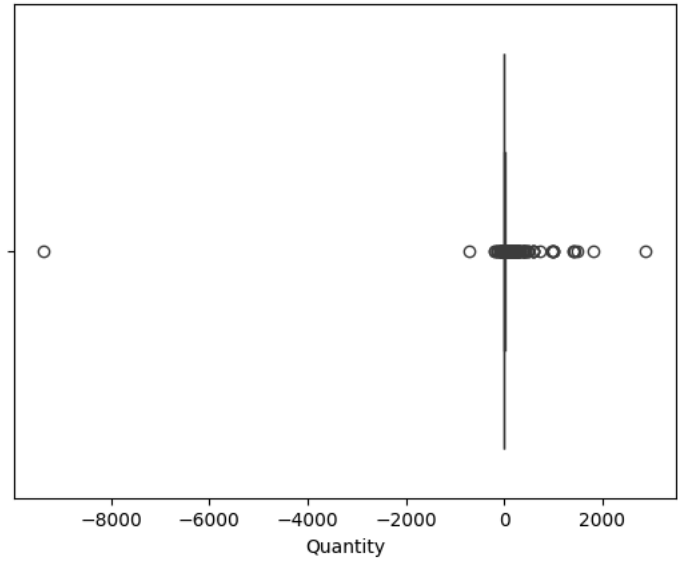
```
sns.histplot(df["Quantity"], bins=50)  
plt.show()
```



```
sns.boxplot(x=df["Revenue"])\nplt.show()
```



```
sns.boxplot(x=df["Quantity"])\nplt.show()
```



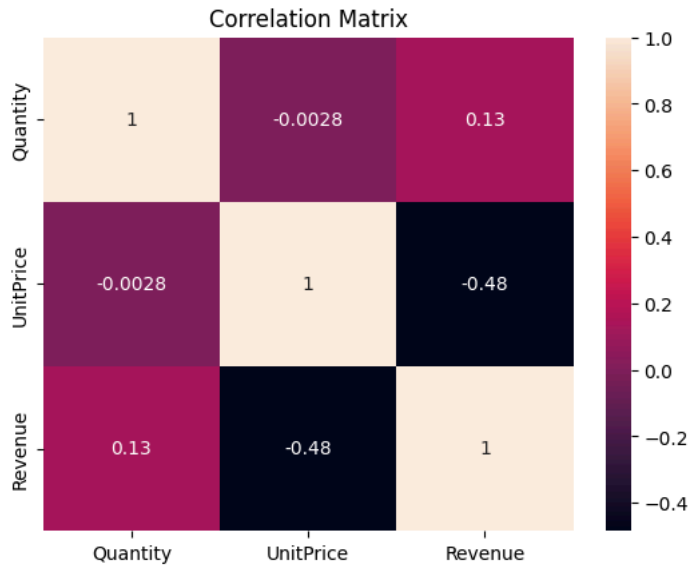
13. CORRELATION ANALYSIS

```
corr = df[["Quantity", "UnitPrice", "Revenue"]].corr()
corr
```

	Quantity	UnitPrice	Revenue
Quantity	1.000000	-0.002815	0.131618
UnitPrice	-0.002815	1.000000	-0.484660
Revenue	0.131618	-0.484660	1.000000

Next steps: [Generate code with corr](#) [New interactive sheet](#)

```
sns.heatmap(corr, annot=True)
plt.title("Correlation Matrix")
plt.show()
```



14. ADVANCED ANALYSIS

i) Top 20% Products Contribution (Pareto Analysis)

```
product_revenue = (
    df.groupby("Description")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)

product_revenue.head(20)
```

	Revenue
Description	
REGENCY CAKESTAND 3 TIER	16883.77
DOTCOM POSTAGE	14558.26
VINTAGE UNION JACK MEMOBOARD	6701.13
PAPER CHAIN KIT 50'S CHRISTMAS	6237.48
WHITE HANGING HEART T-LIGHT HOLDER	6073.86
WOOD BLACK BOARD ANT WHITE FINISH	6032.99
CREAM HEART CARD HOLDER	5280.32
BLACK HEART CARD HOLDER	5085.02
HOT WATER BOTTLE TEA AND SYMPATHY	4932.88
CHILLI LIGHTS	4706.32
RED WOOLLY HOTTIE WHITE HEART.	4609.70
WHITE SKULL HOT WATER BOTTLE	4590.02
ROUND SNACK BOXES SET OF 4 FRUITS	4231.75
BLACK RECORD COVER FRAME	4114.05
CHOCOLATE HOT WATER BOTTLE	3453.62
HAND WARMER BIRD DESIGN	3413.00
ROTATING SILVER ANGELS T-LIGHT HLDR	3288.16
PAPER CHAIN KIT VINTAGE CHRISTMAS	3252.37
ASSORTED COLOUR BIRD ORNAMENT	2774.20
VICTORIAN SEWING BOX LARGE	2724.83

dtype: float64

ii) Cumulative Revenue %

```
cum_rev = (
    product_revenue.cumsum()
    / product_revenue.sum()
) * 100

cum_rev.head(20)
```

	Revenue
Description	
REGENCY CAKESTAND 3 TIER	4.023232
DOTCOM POSTAGE	7.492318
VINTAGE UNION JACK MEMOBOARD	9.089130
PAPER CHAIN KIT 50'S CHRISTMAS	10.575458
WHITE HANGING HEART T-LIGHT HOLDER	12.022798
WOOD BLACK BOARD ANT WHITE FINISH	13.460398
CREAM HEART CARD HOLDER	14.718645
BLACK HEART CARD HOLDER	15.930354
HOT WATER BOTTLE TEA AND SYMPATHY	17.105810
CHILLI LIGHTS	18.227278
RED WOOLLY HOTTIE WHITE HEART.	19.325723
WHITE SKULL HOT WATER BOTTLE	20.419478
ROUND SNACK BOXES SET OF 4 FRUITS	21.427862
BLACK RECORD COVER FRAME	22.408198
CHOCOLATE HOT WATER BOTTLE	23.231161
HAND WARMER BIRD DESIGN	24.044444
ROTATING SILVER ANGELS T-LIGHT HLDR	24.827980
PAPER CHAIN KIT VINTAGE CHRISTMAS	25.602987
ASSORTED COLOUR BIRD ORNAMENT	26.264050
VICTORIAN SEWING BOX LARGE	26.913350

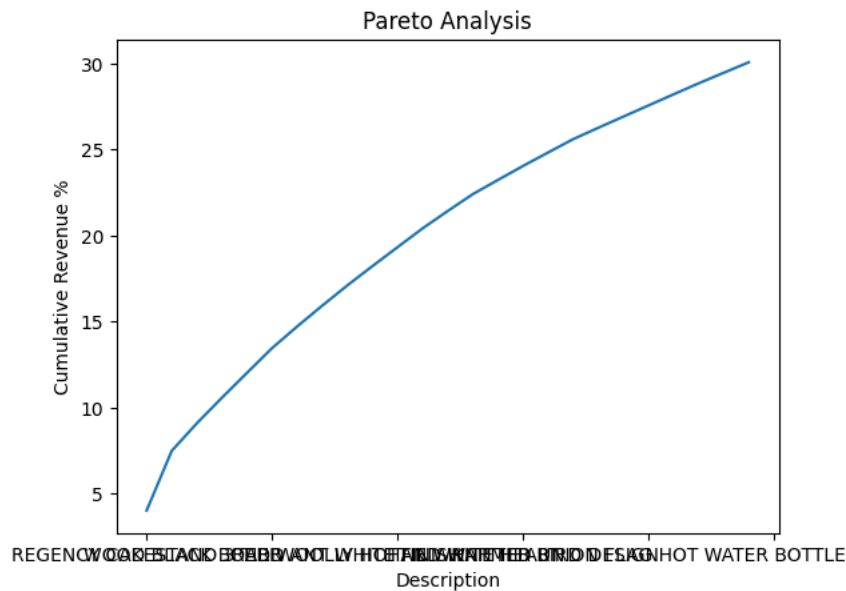
dtype: float64

iii) Pareto Chart

```
cum_rev.head(25).plot()

plt.title("Pareto Analysis")
plt.ylabel("Cumulative Revenue %")

plt.show()
```



15. CUSTOMER SEGMENTATION

i) Customer Revenue

```
customer_revenue = (\n    df.groupby("CustomerID")\n        ["Revenue"]\n        .sum()\n)\n\ncustomer_revenue.head()
```

CustomerID	Revenue
12347.0	711.79
12386.0	258.90
12395.0	346.10
12427.0	303.50
12429.0	1281.50

dtype: float64

ii) Top Customers

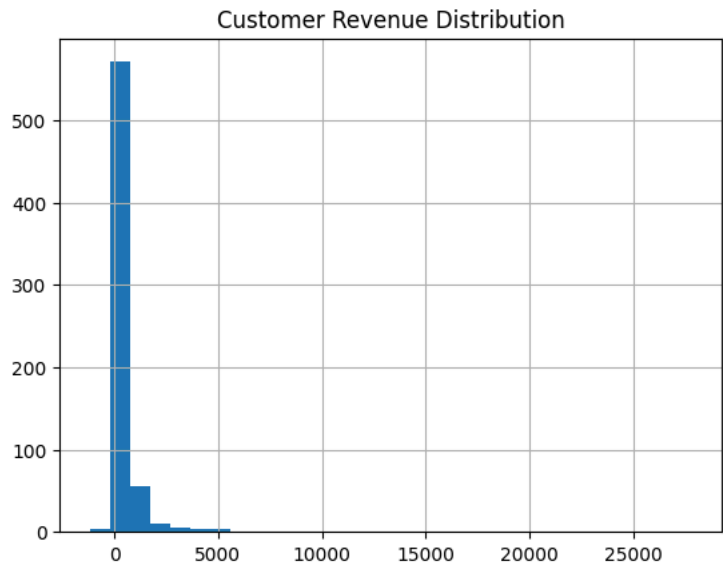
```
customer_revenue.sort_values(\n    ascending=False\n).head(10)
```

CustomerID	Revenue
18102.0	27834.61
15061.0	17344.78
13777.0	6798.16
14911.0	5567.63
17850.0	5391.21
16210.0	4738.54
16029.0	4271.52
12755.0	3794.40
12433.0	3787.12
17381.0	3603.72

dtype: float64

iii) Customer Revenue Distribution

```
customer_revenue.hist(\n    bins=30\n)\n\nplt.title(\n    "Customer Revenue Distribution"\n)\n\nplt.show()
```



16. SEASONALITY ANALYSIS

```
Daily_sales = (
    df.groupby("Day")
    ["Revenue"]
    .sum()
)
Daily_sales
```

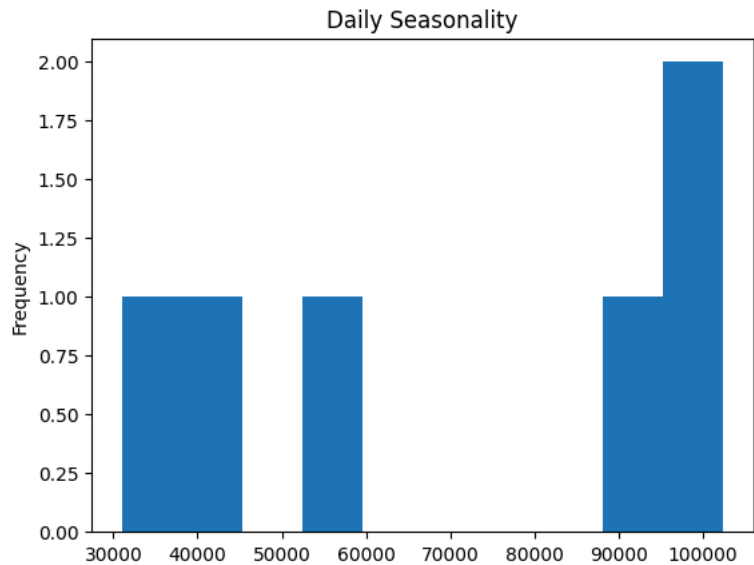
Revenue	
Day	
Friday	88965.46
Monday	53653.87
Sunday	30973.63
Thursday	98582.46
Tuesday	44994.70
Wednesday	102486.78

dtype: float64

```
Daily_sales.plot(
    kind="hist"
)

plt.title(
    "Daily Seasonality"
)

plt.show()
```

17. PEAK SALES MONTH

```
monthly_sales.idxmax()
```

```
'Wednesday'
```

```
monthly_sales.max()
```

```
102486.78
```

18. RREVENUE CONCENTRATION ANALYSIS

```
country_rev = (
    df.groupby("Country")
      ["Revenue"]
      .sum()
)

country_rev.head()
```

Revenue	
Country	
Australia	617.15
Belgium	346.10
Channel Islands	363.53
Denmark	1281.50
EIRE	6431.33

```
dtype: float64
```

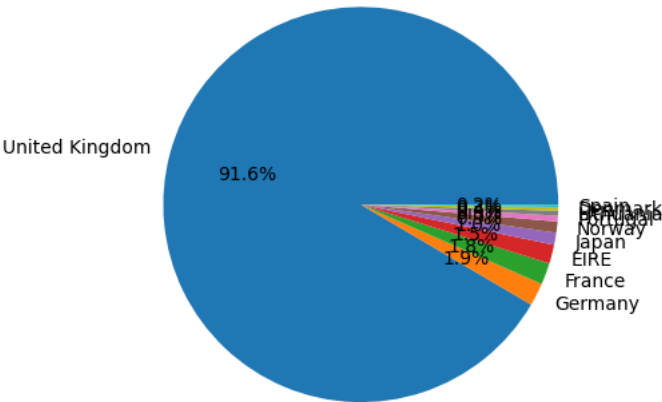
```
country_rev.sort_values(
    ascending=False
).head(10)
```

	Revenue
Country	
United Kingdom	381272.24
Germany	7795.74
France	7335.17
EIRE	6431.33
Japan	4114.48
Norway	3787.12
Portugal	1976.47
Lithuania	1661.06
Denmark	1281.50
Spain	794.72

dtype: float64

```
country_rev.sort_values(
    ascending=False
).head(10).plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.ylabel("")
plt.show()
```



19. OUTLLIER ANALYSIS

```
Q1 = df["Revenue"].quantile(0.25)
Q3 = df["Revenue"].quantile(0.75)
IQR = Q3 - Q1
print(Q1,Q3,IQR)
```

3.36 16.98 13.620000000000001

```
outliers = df[
    (df["Revenue"] <
     (Q1-1.5*IQR))
    |
    (df["Revenue"] >
     (Q3+1.5*IQR))
]
```

outliers.head()

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue	Year	Day
9	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	2010-12-01 08:34:00	1.69	13047.0	United Kingdom	54.08	2010	Wednesday
26	536370	22728	ALARM CLOCK BAKELIKE PINK	24	2010-12-01 08:45:00	3.75	12583.0	France	90.00	2010	Wednesday
27	536370	22727	ALARM CLOCK BAKELIKE RED	24	2010-12-01 08:45:00	3.75	12583.0	France	90.00	2010	Wednesday
28	536370	22726	ALARM CLOCK BAKELIKE GREEN	12	2010-12-01 08:45:00	3.75	12583.0	France	45.00	2010	Wednesday

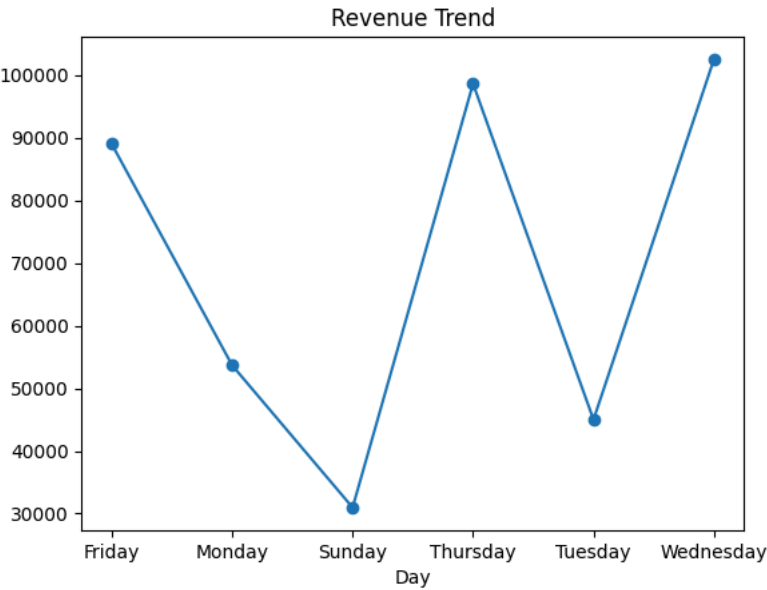
Next steps: [Generate code with outliers](#) [New interactive sheet](#)

20. TREND ANALYSIS

```
monthly_sales.plot(
    marker="o"
)

plt.title(
    "Revenue Trend"
)

plt.show()
```



```
monthly_sales.pct_change()
```

Revenue	
Day	
Friday	NaN
Monday	-0.396913
Sunday	-0.422714
Thursday	2.182787
Tuesday	-0.543583
Wednesday	1.277752

dtype: float64

21.BUSINESS INSIGHTS

```
top_country = (
    df.groupby("Country")
      ["Revenue"]
      .sum()
      .idxmax()
)
```

top_country

'United Kingdom'

```
top_product = (
    df.groupby("Description")
      ["Revenue"]
      .sum()
      .idxmax()
)
```

top_product

'REGENCY CAKESTAND 3 TIER'

```
best_customer = (
    df.groupby("CustomerID")
      ["Revenue"]
      .sum()
      .idxmax()
)
```

best_customer

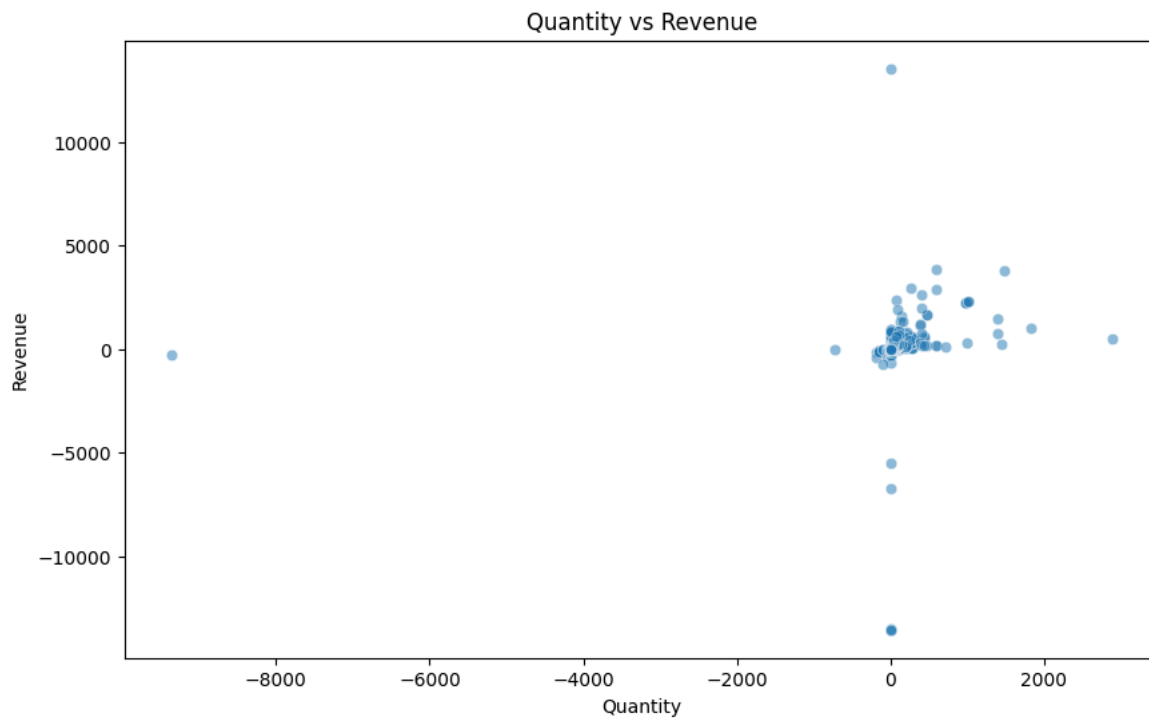
np.float64(18102.0)

22. Quantity Vs Revenue

```
plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x="Quantity",
    y="Revenue",
    alpha=0.5
)

plt.title("Quantity vs Revenue")
plt.show()
```



23. UnitPrice Vs Revenue

```
plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x="UnitPrice",
    y="Revenue",
    alpha=0.5
)

plt.title("Unit Price vs Revenue")
plt.show()
```



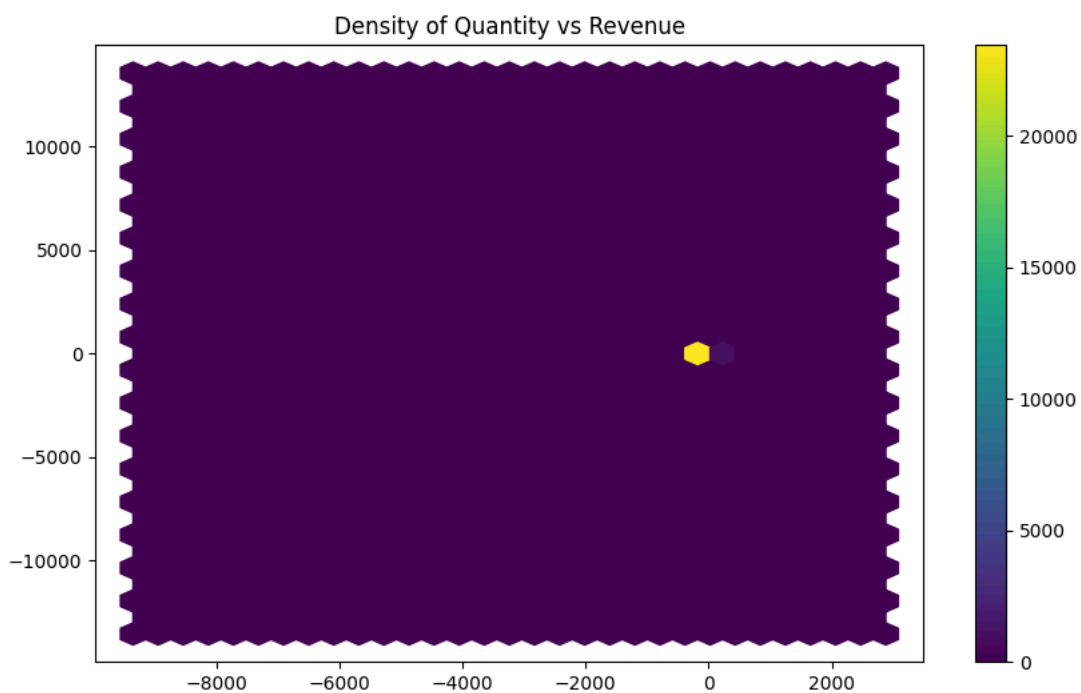
24. HEXBIN PLOT

```
plt.figure(figsize=(10,6))

plt.hexbin(
    df["Quantity"],
    df["Revenue"],
    gridsize=30
)

plt.colorbar()

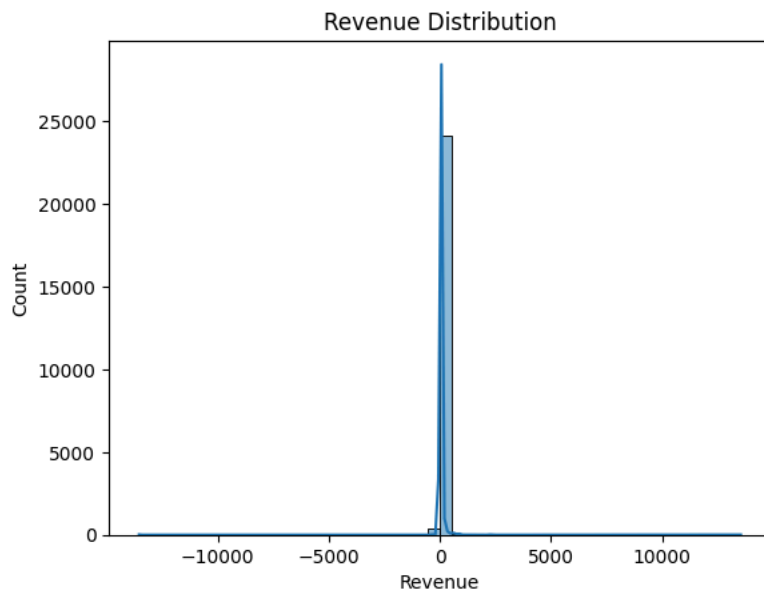
plt.title("Density of Quantity vs Revenue")
plt.show()
```



25.REVENUE HISTOGRAM

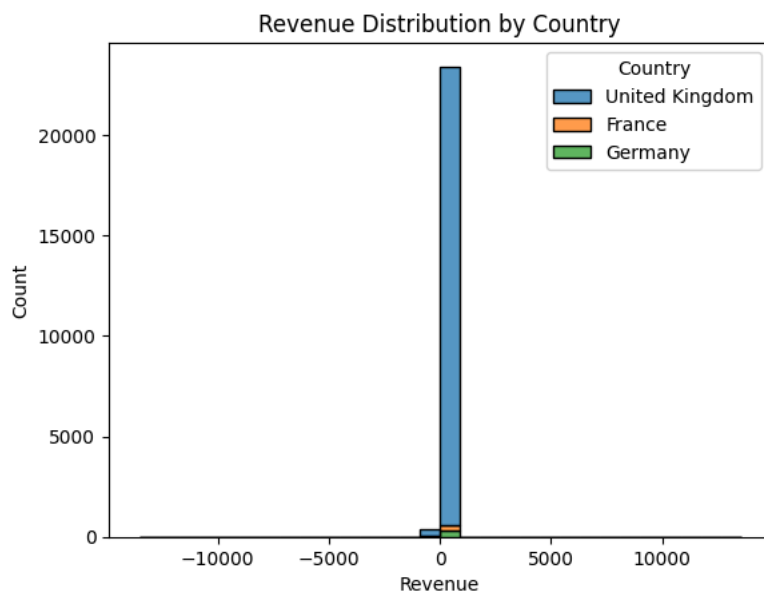
```
sns.histplot(
    df["Revenue"],
    bins=50,
    kde=True
)

plt.title("Revenue Distribution")
plt.show()
```



26. STACKED HISTOGRAM BY COUNTRY

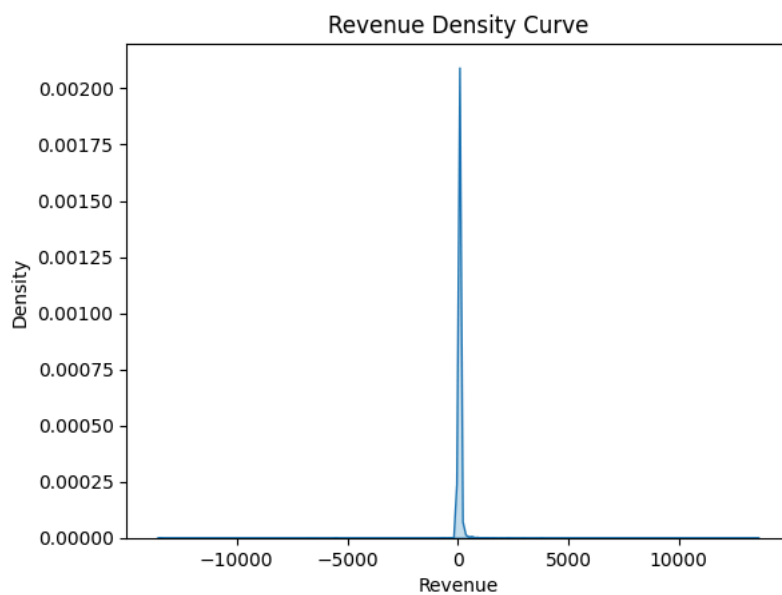
```
top3 = df["Country"].value_counts().head(3).index  
temp = df[df["Country"].isin(top3)]  
  
sns.histplot(  
    data=temp,  
    x="Revenue",  
    hue="Country",  
    multiple="stack",  
    bins=30  
)  
  
plt.title("Revenue Distribution by Country")  
plt.show()
```



27. KDE PLOT

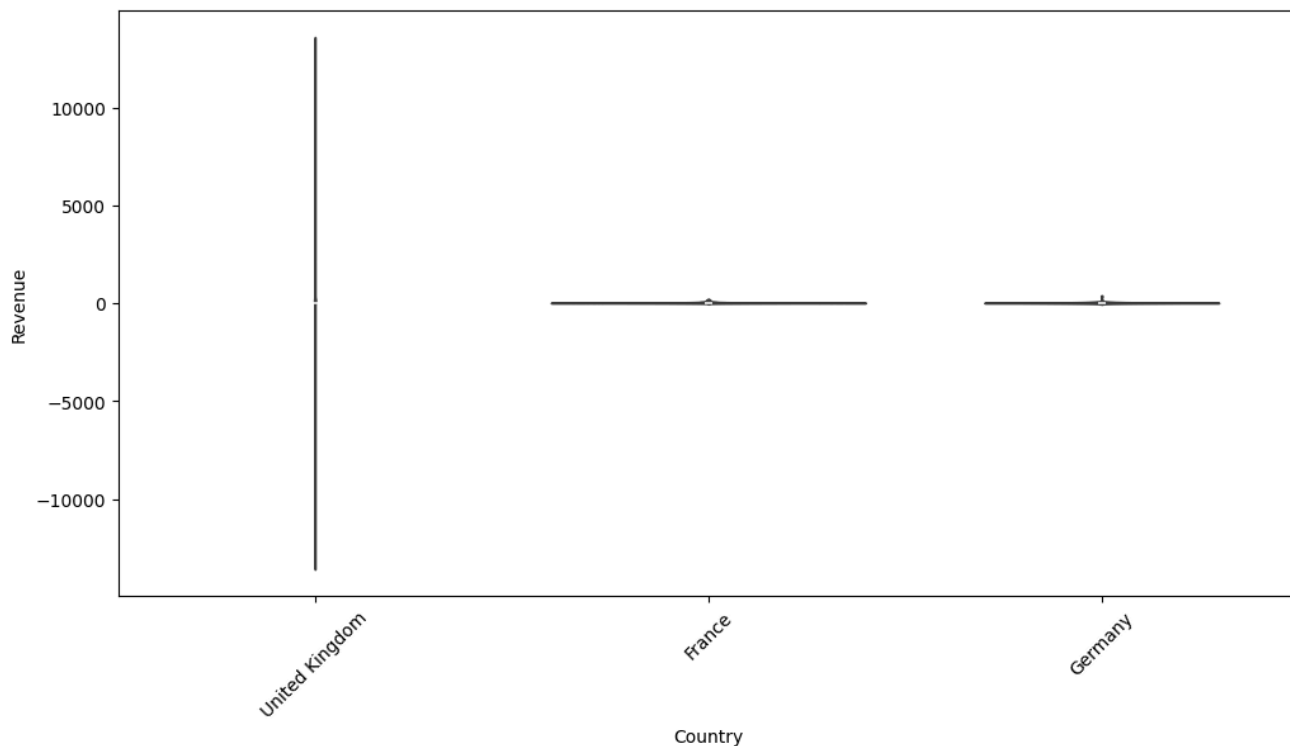
```
sns.kdeplot(  
    df["Revenue"],  
    fill=True  
)
```

```
plt.title("Revenue Density Curve")  
plt.show()
```



28. VIOLIN PLOT

```
plt.figure(figsize=(12,6))  
  
sns.violinplot(  
    data=temp,  
    x="Country",  
    y="Revenue"  
)  
  
plt.xticks(rotation=45)  
plt.show()
```



29. BUSINESS INTELLIGENCE ANALYSIS

i) Top 10 Revenue Products Table

```
df.groupby("Description")["Revenue"] \
.sum() \
.sort_values(ascending=False) \
.head(10)
```

	Revenue
Description	
REGENCY CAKESTAND 3 TIER	16883.77
DOTCOM POSTAGE	14558.26
VINTAGE UNION JACK MEMOBOARD	6701.13
PAPER CHAIN KIT 50'S CHRISTMAS	6237.48
WHITE HANGING HEART T-LIGHT HOLDER	6073.86
WOOD BLACK BOARD ANT WHITE FINISH	6032.99
CREAM HEART CARD HOLDER	5280.32
BLACK HEART CARD HOLDER	5085.02
HOT WATER BOTTLE TEA AND SYMPATHY	4932.88
CHILLI LIGHTS	4706.32

dtype: float64

ii) Bottom 10 Revenue Products

```
df.groupby("Description")["Revenue"] \
.sum() \
.sort_values() \
.head(10)
```

	Revenue
Description	
AMAZON FEE	-39243.08
Discount	-604.91
DOORMAT MERRY CHRISTMAS RED	-182.31
ADVENT CALENDAR GINGHAM SACK	-166.41
CARD HOLDER GINGHAM HEART	-141.79
CARDHOLDER GINGHAM CHRISTMAS TREE	-117.30
Bank Charges	-80.38
SAMPLES	-64.95
ZINC TOP 2 DOOR WOODEN SHELF	-59.80
PINK FOXGLOVE ARTIIFCIAL FLOWER	-30.60